

Random Splits Overstate Machine-Learning Accuracy for 2D-Material Properties Through Structure-Family Overlap: Stiffness, a Property Panel in Two Databases, and Bilayer Binding Energies

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Abstract—Machine-learning models of materials properties are usually validated with random train/test splits. Two-dimensional (2D) materials databases contain many structural relatives—the same prototype with different elements—so a random split may reward recognising a structure type instead of predicting a property. Using 7,258 mechanically stable C2DB monolayers, 141 composition and geometry descriptors (no DFT outputs) and gradient-boosted trees, we find that the mean absolute error (MAE) of the 2D Young’s modulus is 13.8 N/m under random cross-validation and 13.9 N/m when chemical systems are held out, but 19.9 N/m when structure families are held out (ratio 1.44, 95 % cluster-bootstrap interval 1.30–1.62). The family penalty appears for all twelve C2DB properties tested (ratios 1.12–1.92, every interval above 1) and, in a pre-specified replication on JARVIS-DFT 2D, for all ten properties tested (1.03–1.42, every interval above 1), although only seven of the ten exceed our 1.05 materiality threshold, so our pre-specified prediction that at least 80 % would do so failed; three transport properties are near 1. The penalty is not removed by a graph neural network (three seeds; ensemble ratio 1.44 [1.33, 1.55]) or by nested group-aware tuning (1.54 [1.37, 1.77]), and could not be forecast from labels alone (Spearman 0.00 in C2DB, 0.01 in JARVIS, –0.04 pooled; our pre-specified criterion failed each time). For bilayer binding energies from BiDB, grouping by monolayer raises the error 3.9-fold, and tuning does not change this. On 186 JARVIS-DFT materials the stiffness model ranks well (Spearman 0.82) but errs by 29 N/m on unseen chemistry. Of three universal interatomic potentials, only MACE-MPA-0 passes our stiffness gate, and only on C2DB; with a dispersion correction it reproduces BiDB binding energies but not interlayer gaps. We recommend family-grouped evaluation as the default for 2D-materials benchmarks. No bilayer stiffness is DFT-validated here.

Index Terms—2D materials, Young’s modulus, data leakage, grouped cross-validation, gradient boosting, graph neural network, universal interatomic potentials, bilayers



1 INTRODUCTION

Two-dimensional (2D) crystals are attractive for flexible electronics, membranes and composites, where in-plane stiffness decides whether a sheet survives handling and strain. Thousands of hypothetical 2D crystals can be generated, but measuring each is impossible and computing its stiffness from first principles is expensive: every material needs a relaxed reference cell plus several strained density-functional-theory (DFT) calculations. Machine-learning (ML) surrogates trained on existing DFT results promise a cheaper estimate and have been proposed for 2D materials [1], [2].

How reliable are such surrogates on *new* materials? The common practice is a random train/test split. 2D databases contain many polymorphs and prototype relatives (MoS₂, MoSe₂ and WS₂ share one structure type), so a random split places close relatives of almost every test material in the training set, and the reported error partly measures recognition of a familiar structure. That random cross-validation overestimates performance, and that group- or cluster-wise hold-out is harder, is well established for materials data [3], [4], [5], [6], [7]; we do not claim it as new. The open questions for 2D layered materials are *which* overlap matters—chemistry or structure prototype—whether the effect is specific to one property or database, whether a better model class or tuning removes it, and whether it can be forecast cheaply.

We therefore treat *evaluation* as the object of study and use deliberately plain, DFT-free descriptors and standard regressors. Our contributions, in decreasing order of evidential strength, are:

- In C2DB the inflation of the 2D Young’s modulus accuracy comes from **structure-family** overlap, not chemistry: holding out families raises the MAE 1.44× [1.30, 1.62] whereas holding out chemical systems changes it by 1 % (Section 4.3).
- The same family penalty appears for **all 22** property–database pairs we tested (12 in C2DB, 10 in a pre-specified JARVIS-DFT replication), with cluster-bootstrap intervals; it exceeds our 1.05 materiality threshold for 19 of the 22, and a pre-specified prediction that it would for 80 % of the JARVIS properties **failed** (Section 4.5). It **survives** a three-seed graph neural network and nested group-aware tuning, also for bilayer binding energy (Section 4.6).
- A label-only statistic computed under random splits **does not forecast** how much skill a property loses on unseen families, in either database or pooled; we report this pre-specified negative result (Section 4.7).
- For BiDB bilayers, monolayer-level leakage dominates: grouping by monolayer raises the binding-energy error 3.9-fold (Section 4.8).
- An external JARVIS test and a gated calibration of universal interatomic potentials that quantify what can and cannot yet be trusted (Sections 4.9–4.11).

We do **not** claim a new model or state-of-the-art accuracy, and we state what is not done: no bilayer *stiffness* model or DFT-validated bilayer stiffness, no mechanism for the family effect, no prospective test on an independent dataset, and no feature-level attribution (SHAP, permutation importance).

2 BACKGROUND AND RELATED WORK

2.1 Mechanical descriptors of a 2D sheet

For small in-plane strain a sheet obeys a linear stress–strain law in Voigt notation with a 3×3 stiffness matrix over (xx, yy, xy). Because a sheet has no well-defined thickness, the stress is integrated over the cell height L_z (vacuum included) and the constants are quoted in N/m (force per unit edge length), not GPa:

$$\sigma_i^{2D} = \sigma_i^{3D} L_z, \quad C_{ij} = \frac{\partial \sigma_i^{2D}}{\partial \varepsilon_j}. \quad (1)$$

The in-plane Young’s modulus and Poisson ratio follow from the xx, yy block:

$$Y_{2D} = \frac{C_{11}C_{22} - C_{12}^2}{C_{22}}, \quad \nu = \frac{C_{12}}{C_{22}}. \quad (2)$$

N/m scales with the number of layers whereas GPa does not, so a monolayer–bilayer comparison must state whether a value is per layer or for the whole slab. All stiffness values here are single-layer or explicitly *total* for the slab.

2.2 Traditional routes to stiffness

Experiment. Suspended sheets are measured by nanoindentation or from the nonlinear response of drum resonators [8]. Reported values carry large spreads: a review table lists graphene at 1000 ± 100 GPa and monolayer MoS_2 at 270 ± 100 GPa [9]. Samples must first be made, which is impractical for hypothetical materials.

Classical molecular dynamics. Atomistic nanoindentation or tensile simulations with empirical potentials are fast [10], [11] but the potential must be parameterised per chemistry, so results transfer poorly.

DFT finite-strain method. The reference route, used to build C2DB, relaxes the sheet, applies ± 1 % strains in xx, yy and xy, relaxes the ions and reads the stress. With strain amplitude δ and $\kappa = 16.0218 \text{ N m}^{-1} \text{ per eV } \text{\AA}^{-2}$,

$$C_{ij} = \kappa L_z \frac{\sigma_i(+\delta) - \sigma_i(-\delta)}{2\delta}, \quad \delta = 0.01. \quad (3)$$

This needs one reference relaxation and six strained relaxations per material: accurate and general, but slow.

2.3 Machine learning and how it was validated

Table I summarises the seven papers on our reading list (a reading list, not a systematic review).

TABLE I — PRIOR 2D-MECHANICS WORK REVIEWED AND ITS SPLIT

Study	Data / model	Validation split
Venturi et al. [1]	C2DB ($\approx 3,500$); 100-model CGCNN ensemble; formation energy, gap, stiffness constants	Random 70 % per model; metrics over all of C2DB, training points included
Fronzi et al. [2]	Bilayers; 282 interlayer-energy, 226 C_{33} labels; Bayesian NN	K-means cluster-stratified 75/25 (in-domain)
Malakar et al. [11]	1,440 MD runs, four TMDs; FFNN, LSTM	Ratio not stated; test from the same four materials
Maurizi et al. [12]	FE composites and lattices; GNN	Random 90/10
Elder; Davidovikj; Zhang [10], [8], [9]	MD / experiment / review (no ML)	—

All ML studies in Table I use random or in-domain test sets; none evaluates on held-out chemical systems or structure families. Closest to our bilayer target, Fronzi et al. predict interlayer energy and the out-of-plane constant C_{33} (test $R^2 \approx 0.80$) [2]. In the wider materials-informatics literature, leave-one-cluster-out evaluation [3], redundancy control [5], [6], standardised chemistry- and structure-based hold-outs [7] and benchmark suites [13] document the same optimism, and a recent BiDB study already splits at the monolayer level [14]. Our added value is narrower: a quantified comparison, for 2D mechanical and interlayer targets, of *which* grouping removes the optimism, with interval estimates and tests of model class, tuning and a cheap forecast.

3 METHOD

3.1 Data and mechanical-stability filtering

The training source is C2DB [15], [16] (GPAW [17], PBE [18]); the external source is the 2D subset of JARVIS-DFT [19], [20] (VASP [21], OptB88vdW [22]); bilayer labels come from BiDB [23] (PBE-D3). From each stiffness file we form the symmetrised 3×3 (xx, yy, xy) matrix $\mathbf{C}$ and keep a material only if it is finite, positive definite and nearly symmetric:

$$\begin{aligned} \lambda_{\min}(\tfrac{1}{2}(\mathbf{C} + \mathbf{C}^T)) &> 0, \\ \frac{\max_{ij} |C_{ij} - C_{ji}|}{\max_{ij} |C_{ij}|} &\leq 0.10. \end{aligned} \quad (4)$$

A sheet that does not resist stretching in every direction is mechanically unstable, and large asymmetry flags a noisy calculation. Stability is derived from the tensor, never from a database label. For JARVIS only the xx, yy block is used (its shear entries did not reproduce known MoS_2 values) and corrupt tensors (NaN or denormal numbers) are dropped. For the Poisson target rows with $|v| > 1$ are removed.

BiDB provides 10,192 homobilayers of 992 monolayers (109 structure families). We use the rigid vertical-scan binding energy ($\text{meV}/\text{\AA}^2$ per interface, positive = bound) and the interlayer gap ($\text{\AA}$); definitions were established from the files and published summaries. A label is valid if it is positive and below the upper tail $Q3 + 3 \text{ IQR}$ of its $\log_{10}$ distribution. A uid map links 982 of 992 monolayers one-to-one to C2DB (verified at formula level only).

3.2 Groups used for splitting

The *chemical system* is the sorted element set ($\text{MoS}_2 \rightarrow \text{Mo-S}$). The *structure family* joins the anonymised formula (element identities hidden, so MoS_2 and WSe_2 both become AB_2) with the layer-group number N_{lg} :

$$\begin{aligned} k_{\text{chem}} &= \text{sorted}(\{\text{elements}\}), \\ k_{\text{fam}} &= \text{anon}(\text{formula}) \parallel \lg N_{\text{lg}}. \end{aligned} \quad (5)$$

The clean stiffness set has 3,806 chemical systems (2,572 single-member) and 676 families (largest 432 rows). Polymorphs are kept as separate rows, which is why family grouping matters. For BiDB the groups are the monolayer and the family (anonymous stoichiometry plus monolayer layer group).

3.3 Descriptors

Each C2DB material is described by 141 numbers: **132 composition** descriptors (Magpie preset [24], matminer [25], pymatgen [26]) and **9 structural** ones (lattice lengths a , b , angle γ , area per atom, atom count, slab thickness, n_{layers} , space-group number, number of elements). In matminer, Magpie summarises 22 elemental properties with six composition-weighted statistics each ($22 \times 6 = 132$); for a property p and atomic fractions x_i ,

$$\bar{p} = \sum_i x_i p_i, \quad \langle |\Delta p| \rangle = \sum_i x_i |p_i - \bar{p}|, \quad (6)$$

plus minimum, maximum, range and mode. Because Magpie depends only on the reduced formula, polymorphs share identical composition descriptors; geometry separates them. In the monolayer data n_{layers} is constant. DFT outputs (energy above hull, formation energy, band gap, stability labels) are excluded for the stiffness task: they need a calculation, so a model using them could not score a genuinely new material. For BiDB the monolayer feature sets are *mono_basic* (140: Magpie, geometry, monolayer layer group), *mono_stiffness* (147: adds C2DB monolayer stiffness through the uid map) and, as an extension, *mono_stiffness_stacking* (158: adds stacking descriptors from the uid). Because the first two describe only the monolayer, every stacking of a monolayer receives the same prediction.

3.4 Models

Stiffness is right-skewed (median 47.9, mean 66.6, maximum 713.6 N/m), so models are fitted to $\ln Y_{2D}$ and mapped back; the Poisson ratio is not transformed. Three regressors from scikit-learn [27] and LightGBM [28] use fixed settings: ridge regression [29] with internal-CV penalty,

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \|\mathbf{z} - \mathbf{X}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_2^2, \quad (7)$$

a random forest [30] (300 trees, one third of features per split); and LightGBM (500 trees, learning rate 0.05, 31 leaves, 80 % row and 50 % column subsampling), an additive model

$$F_M(\mathbf{x}) = \sum_{m=1}^M \eta f_m(\mathbf{x}), \quad M = 500, \quad \eta = 0.05. \quad (8)$$

To test whether a structure-aware model narrows the gap we trained a CGCNN-type network [31] in PyTorch on the same rows and the *identical* folds (seed 42): three gated graph-convolution layers,

$$\mathbf{v}_i^{(t+1)} = \mathbf{v}_i^{(t)} + \sum_{j \in \mathcal{N}(i)} \sigma(\mathbf{z}_{ij} \mathbf{W}_f + \mathbf{b}_f) \odot g(\mathbf{z}_{ij} \mathbf{W}_s + \mathbf{b}_s), \quad (9)$$

with $\mathbf{z}_{ij} = \mathbf{v}_i \oplus \mathbf{v}_j \oplus \mathbf{u}_{ij}$, 64 hidden units, mean pooling and a two-layer read-out. Node inputs are ten standardised elemental properties; edges join all neighbours within 6 Å (periodic in-plane) with 40 Gaussian distance functions. Training used L1 loss, Adam, up to 200 epochs, and early stopping on a validation subset drawn *with the same grouping rule as the outer split*, so early stopping never sees test-family information. The network was trained with three seeds (42–44) and also evaluated as a three-seed prediction ensemble; no hyper-parameter search. Separately, LightGBM was tuned by random search (16 configurations plus the default for Y_{2D} ; 12 plus the default for the BiDB binding energy) with three-fold *inner* cross-validation

that matches the outer grouping, then refitted and scored on the outer test fold.

3.5 Evaluation protocol

Five-fold cross-validation is run under four schemes: **random** (comparison only), **cluster** (K-means, 20 clusters on standardised Magpie features, stratified so every cluster appears in train and test, after Fronzi et al. [2]), **chemical-system** and **structure-family** (monolayer and family for BiDB). Grouped schemes use stratified group K-fold on ten target-quantile bins, which balances the target distribution while keeping each group in one fold. Overlap is measured explicitly: for a key k , the fraction of test rows whose key also occurs in training,

$$L_k = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \mathbf{1}[k_i \in \{k_j\}_{j \in \mathcal{R}}], \quad (10)$$

with test fold $\mathcal{T}$ and training rows $\mathcal{R}$. Metrics are the MAE, the coefficient of determination (also in log space) and the ratio of grouped to random error:

$$\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i|, \quad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad (11)$$

$$\mathcal{R}_s = \frac{\text{MAE}_s}{\text{MAE}_{\text{random}}}.$$

Seeds 42–46 are used for the stiffness case study and 42–44 for the panel. Seed standard deviations hold the dataset fixed and therefore understate uncertainty, so all headline intervals are **cluster bootstraps**: the clustering unit (structure family; monolayer for BiDB) is resampled with replacement (2,000 resamples), pooled out-of-fold metrics are recomputed, and 2.5th–97.5th percentiles are reported; paired differences use the same resample. Null baselines (global, chemical-system and family means) show how much skill group identity alone provides.

3.6 Property panel and label-only forecast

To test generality we built a panel of C2DB numeric fields by fixed rules: at least 3,000 labels; not an identifier, symmetry integer or purely geometric quantity; not degenerate (<50 % of values at the mode); one representative per property family (PBE and HSE gaps are distinct). Twelve targets qualify: formation energy (hform), energy above hull (ehull), PBE and HSE gaps, vacuum level (evac), Fermi level (efermi), valence-band maximum (vbm), electronic polarisability (alphax_el), plasma frequency (plasmafrequency_x), conduction-band effective mass (emass_cbm), Y_{2D} and v . Targets are log-transformed if all values are positive with max/min > 100. The same 141 features and default LightGBM are used for all. With skill measured against the mean predictor (MAE_0), we define skill and skill retention,

$$S_s = 1 - \frac{\text{MAE}_s}{\text{MAE}_0}, \quad R_s = \frac{S_s}{S_{\text{random}}}, \quad (12)$$

and a label-only diagnostic computed with no features: the skill of a family-mean predictor under random splits,

$$D_{\text{fam}} = 1 - \frac{\text{MAE}(\text{family-mean predictor})}{\text{MAE}(\text{global-mean predictor})}. \quad (13)$$

The pre-specified hypotheses were: H1, every ratio >1 (with ≥ 1.05 reported) and every $R_{\text{fam}} < 1$; H2 (primary), Spearman

$\rho(D_{\text{fam}}, R_{\text{fam}}) \leq -0.6$ with a 95 % interval below 0; H3, a leave-one-property-out linear fit beats the mean (error ratio <0.8); H4, the chemical-system analogue; H5, D_{fam} associates with R_{fam} at least as strongly as the intra-class correlation or simple covariates. H2 was to be called not predictive if $\rho > -0.3$ or the error ratio ≥ 1 . These hypotheses and the graph-network and tuning designs were written in dated design files before the corresponding models were fitted and their thresholds were not changed afterwards; the files were committed together with the results, so this order is recorded by the authors and cannot be verified from version control.

Prospective replication on JARVIS-DFT 2D. To test whether the effect belongs to C2DB or to the splitting problem, the design, target-selection rules and five predictions (P1–P5) were written before any JARVIS panel model was fitted. All 1,103 JARVIS monolayers are used with the same 141 features and protocol (three seeds). JARVIS has no layer group, so the family is the anonymous formula plus the space-group number, a documented deviation from C2DB’s definition. Ten properties met the same rules (formation energy, OptB88vdW gap, ϵ_x , effective mass, spillage, exfoliation energy, n-Seebeck, n-power factor, n-conductivity, n- κ). The predictions were: P1, ratio >1.05 for ≥ 80 % of targets with the interval above 1 for ≥ 70 %; P2, chemical-system ratio <1.15 for ≥ 70 %; P3, H2 fails to replicate ($|\rho| < 0.5$); P4, the pooled 22-property test also fails; P5, formation energy and the band gap show the penalty in both databases. If P1 failed, the claim was to be softened from “general” to “present in the properties tested”.

3.7 External validation on JARVIS

The stiffness model is trained on all clean C2DB rows and applied to 186 clean JARVIS rows with no JARVIS label in training. Rows are matched to C2DB by formula, atom count and lattice constants within 3 % (a heuristic), giving 106 *matched* rows and 55 rows whose chemical system is absent from C2DB (*unseen chemistry*). Intervals are 95 % bootstrap intervals [32] over JARVIS rows; ranking is measured by the Spearman coefficient [33],

$$\rho = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)}, \quad (14)$$

$$d_i = \text{rank}(y_i) - \text{rank}(\hat{y}_i).$$

(formula valid without ties). The agreement of the two databases on the 106 shared materials sets a noise floor for cross-database comparison.

3.8 Universal interatomic potentials

MACE-MP-0 (“medium”), MACE-MPA-0 (“medium-mpa-0”) [34] and CHGNet 0.3.0 [35] are driven through ASE [36] to reproduce Eq. (3): ions and in-plane cell relaxed ($F_{\text{max}} = 0.005$ eV/Å), then ± 1 % strains with ion relaxation. Each run first checks that the analytic stress equals a finite-difference energy derivative (ratio 0.97–1.03 required). Cells are fully periodic with ≥ 15 Å vacuum (a non-periodic vacuum axis gave a wrong stress in an early test). The benchmark has 150 C2DB monolayers (one member from each of 150 randomly drawn families of a 3,380-monolayer pool: non-magnetic, $E_{\text{hull}} \leq 0.1$ eV/atom, ≤ 12 atoms) and 184 JARVIS rows (≤ 24 atoms). The **stiffness gate** was fixed beforehand:

$$\rho \geq 0.90 \quad \wedge \quad \text{MAE} \leq 1.5 \text{ MAE}_{\text{DB}} = 12.93 \text{ N/m}, \quad (15)$$

$$\text{MAE}_{\text{DB}} = 8.62 \text{ N/m (C2DB vs JARVIS labels)}.$$

A **fixed-cell diagnostic** reruns MACE-MPA-0 on the unchanged C2DB cell to test whether geometry explains its error. For **interlayer binding**, MACE-MPA-0 with D3(BJ) dispersion [37] scans the vertical gap of each rigid BiDB bilayer (30 non-magnetic monolayers, 307 bilayers) like-for-like with the DFT labels, with

$$E_b = -\frac{E_{\text{bilayer}} - 2E_{\text{mono}}}{A} \times 10^3 \text{ [meV/Å}^2\text{]}. \quad (16)$$

The pre-specified BiDB criteria were: D1 (gap) median $|\Delta d| \leq 0.10$ Å and 90th percentile ≤ 0.30 Å; E1 Spearman ≥ 0.80 ; E2 median ratio in $[0.8, 1.25]$, ≥ 70 % within ± 30 % and ≥ 90 % within a factor 2; E3 failures ≤ 10 %; S1 mean within-monolayer Spearman ≥ 0.60 with the BiDB-best stacking in the top two for ≥ 60 % of monolayers. Finally, 60 homobilayers (best and worst reference stacking of each of the 30 monolayers) were strained like monolayers, and the descriptive ratio

$$R = \frac{Y_{2D}^{\text{total}}(\text{bilayer})}{Y_{2D}(\text{monolayer})} \quad (17)$$

was recorded (≈ 2 for additive layers). No gate was attached to it.

4 EXPERIMENTAL RESULTS

4.1 Dataset statistics

Of 16,905 C2DB entries, 8,462 have a stiffness file; 988 fail positive-definiteness and a further 216 the asymmetry test, leaving 7,258 (14.2 % of labelled rows removed; Fig. 2). For JARVIS, 18 of 229 tensors are corrupt and 25 more are unstable in the xx, yy block, leaving 186. For BiDB, 9,993 binding-energy and 10,189 gap labels are valid (Table II). The JARVIS panel uses all 1,103 JARVIS monolayers (no stiffness required; 603–1,103 labels per property).

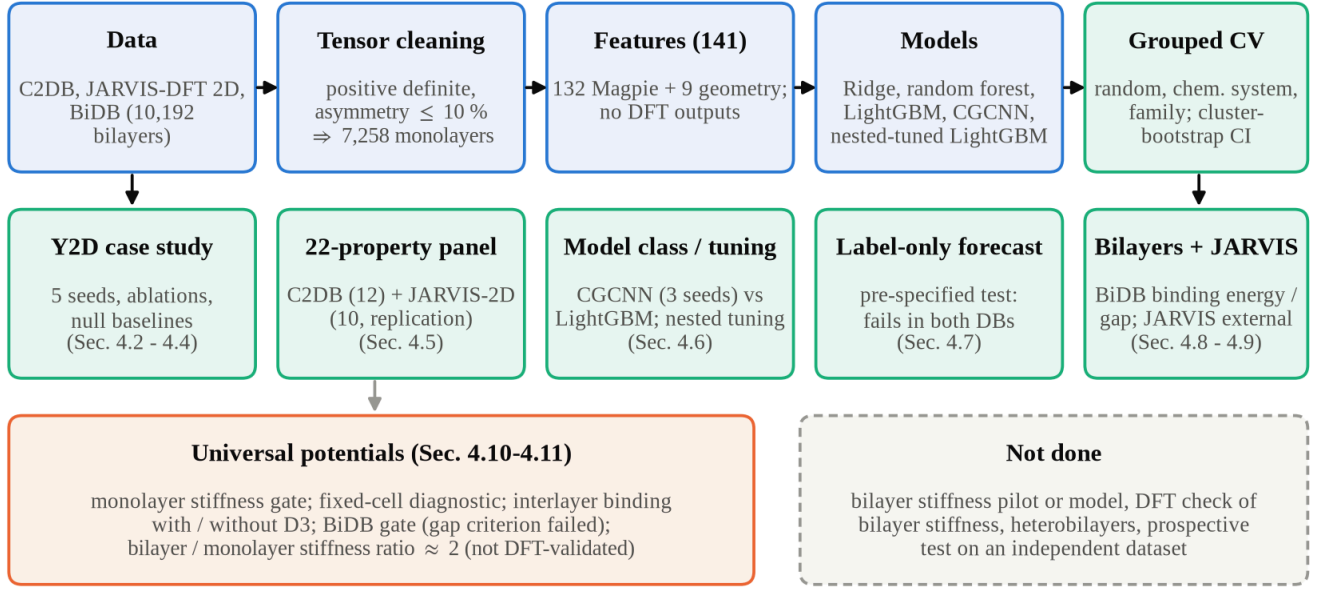


Fig. 1. Workflow. Blue: data and model construction. Green: evaluation experiments completed. Orange: universal-potential checks. The dashed grey box lists work that has *not* been done.

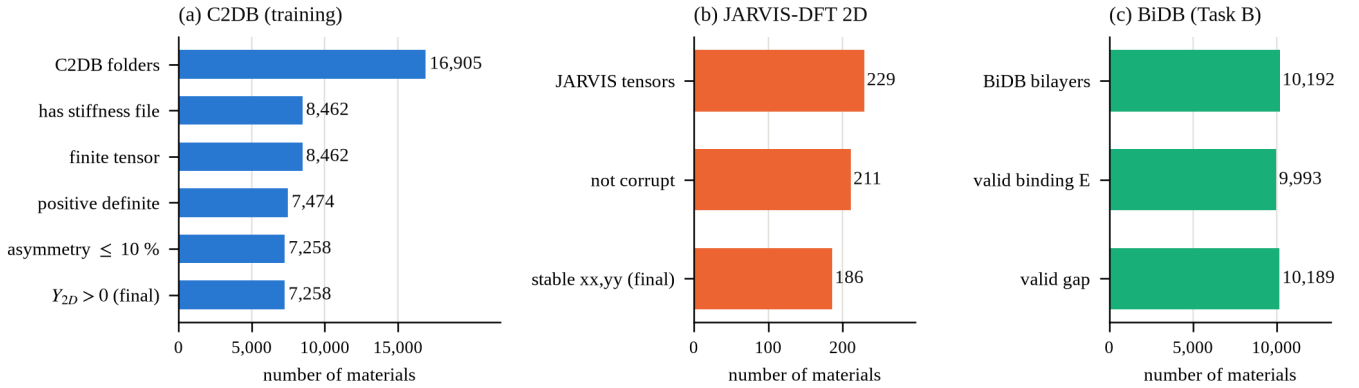


Fig. 2. Data attrition. (a) C2DB: from all entries to the 7,258 mechanically stable monolayers. (b) JARVIS-DFT 2D: from 229 tensors to the 186 used as the external set. (c) BiDB: valid labels of the 10,192 homobilayers.

TABLE II — CLEAN DATASETS

Item	C2DB	JARVIS	BiDB
Rows used	7,258	186 (1,103 panel)	9,993 / 10,189
Monolayers / families	7,258 / 676	—	992 / 109
Chemical systems	3,806	—	—
Median label	47.9 N/m	71.1 N/m	—
Role	train / CV	external	Task B, checks

4.2 Baselines (seed 42)

The mean predictor has MAE 42.4 N/m. Ridge halves this (22.5), trees reduce it by about a third again (14.7 for the forest) and LightGBM is best in every scheme, but only about 1 N/m better than the forest (Table III): the non-linear dependence matters more than the choice between tree ensembles. The ridge family-split R^2 of -11.0 is a back-transformation artefact of one fold (its log-space R^2 is 0.36). The Poisson ratio is predicted poorly (LightGBM $R^2 = 0.39$ random, 0.14 family).

TABLE III — BASELINES, SEED 42, MAE (N/m) / R^2

Model	Random	Cluster	Chem. sys.	Family
Ridge	22.5 / 0.69	22.6 / 0.69	22.8 / 0.67	31.1 / -11.0
Random forest	14.7 / 0.85	14.7 / 0.84	14.8 / 0.83	20.9 / 0.67
LightGBM	13.7 / 0.87	14.0 / 0.87	13.9 / 0.86	19.8 / 0.70
Poisson, LightGBM	0.111 / 0.39	0.111 / 0.39	0.109 / 0.42	0.140 / 0.14

4.3 Random versus grouped validation (Y_{2D})

Overlap. Fig. 3 shows why the schemes differ. Under random splitting 96 % of test rows have their structure family in training and 61 % their chemical system. The chemical-system split drives chemical overlap to zero but leaves 96 % family overlap; only the family split removes it (keeping 59 % chemical overlap).

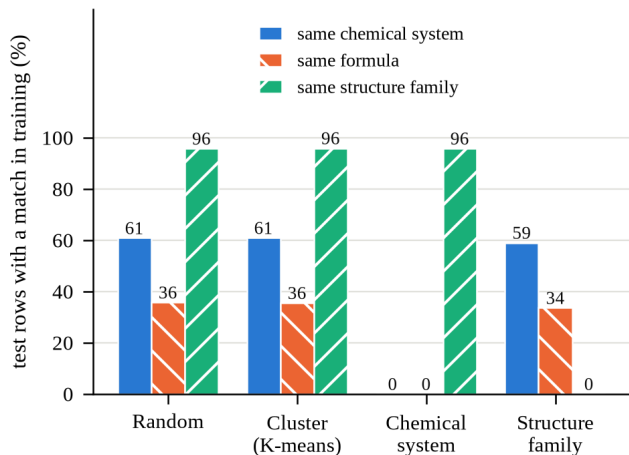


Fig. 3. Measured leakage, Eq. (10): fraction of test rows whose chemical system, formula or structure family also occurs in training.

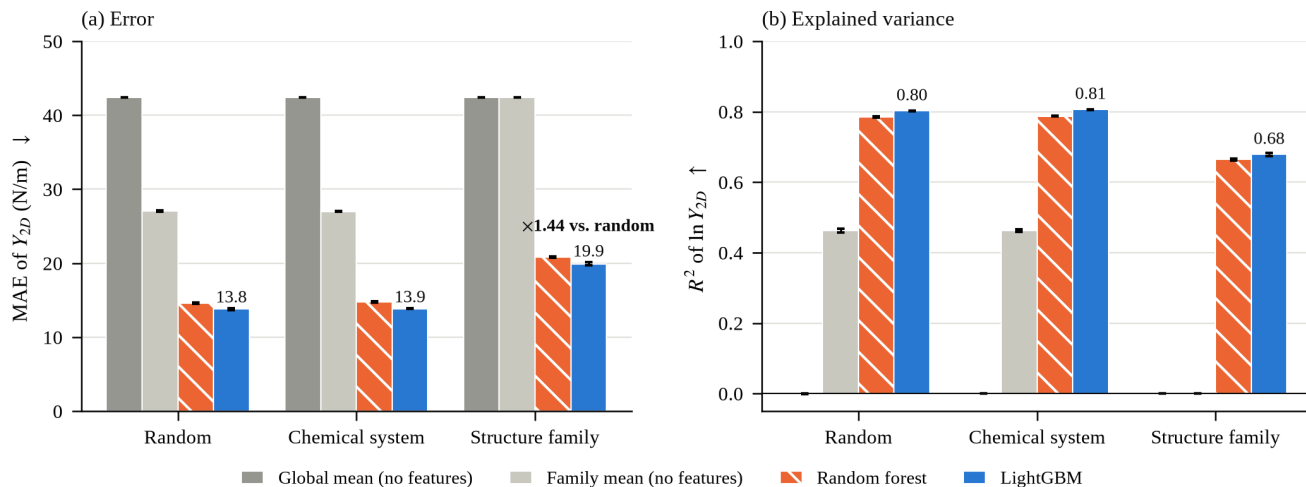


Fig. 4. Five-seed cross-validation of Y_{2D} (error bars: s.d. over seeds). (a) MAE. (b) R^2 of $\ln Y_{2D}$. The no-feature family-mean baseline explains 46 % of the variance when family members are shared and none when they are not.

TABLE IV — Y_{2D} , FIVE SEEDS, MAE (N/m)

Model	Random	Chem.	Family	Fam./Rand.
Global mean	42.4	42.4	42.4	1.00
Family mean	27.0	27.0	42.4	1.57
Chem.-sys. mean	34.5	42.4	34.3	0.99
Random forest	14.6	14.7	20.8	1.43
LightGBM	13.8	13.9	19.9	1.44

Seed s.d. 0.1–0.2 for learned models; it understates uncertainty from the fixed sample (Section 3.5).

What this shows. The random-split score is optimistic for unfamiliar structure types by about 44 % in error, while the model stays well above the no-feature baselines (19.9 vs

Accuracy. Over five seeds LightGBM attains MAE 13.8 ± 0.1 (random), 13.9 ± 0.1 (chemical system) and 19.9 ± 0.2 N/m (family) (Fig. 4, Table IV); log-space R^2 falls from 0.80 to 0.68. The family/random ratio is 1.44 ± 0.02 over seeds, with a seed-42 cluster-bootstrap interval of [1.30, 1.62], and 1.43 for the random forest; the chemical-system ratio is 1.01 [0.99, 1.04]. The cluster scheme gives 1.02 (seed 42), as expected of an in-domain split. Pooled out-of-fold R^2 drops from 0.874 [0.828, 0.905] to 0.740 [0.602, 0.828].

42.4 N/m). The chemical-system split, which one might expect to be harder, is not: removing the 61 % chemical overlap changes nothing because 96 % structural overlap remains. **Is the gap just missing features?** (i) Giving the model layer group and anonymous formula directly leaves the ratio at 1.44–1.47 (Section 4.4). (ii) The family-mean baseline falls from $R^2(\ln) = 0.46$ to 0.00 once the family is unseen, i.e. identity alone explains the random-split gain but nothing beyond it. (iii) Composition-only LightGBM still reaches $R^2(\ln) = 0.50$ on unseen families, so transferable chemical signal exists; the remaining gap is the cost of extrapolating to new prototypes.

4.4 Feature ablation

Fig. 5 removes groups of descriptors. Composition only (19.2/25.5 N/m, random/family) and structure only (19.8/26.6) perform almost identically, and combining them lowers the error by about 28 % (13.8/19.9): the two views are complementary though partly redundant. The space-group number alone is nearly uninformative (38.0/44.3, worse than the 42.4 mean predictor under family CV), and removing it from the full model costs only 0.3 N/m, so the structural signal is carried by geometry.

One-hot layer-group and anonymous-formula columns alone recover the family-mean baseline under random CV ($R^2(\ln)$ 0.455 vs 0.463) and nothing under family CV (0.06); added to the full set they change the MAE by -0.2 to $+0.1$ N/m, within one to three seed standard deviations. Ablation removes *groups* of columns with one model, and shows predictive contribution, not physical causation; no SHAP or permutation analysis was run.

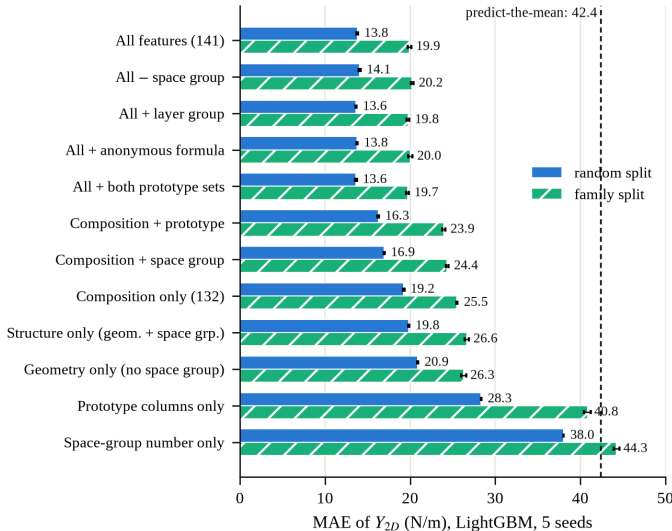


Fig. 5. Group-level ablation (LightGBM, five seeds; bars: random split, hatched: family split; error bars: s.d. over seeds). Dashed line: predict-the-mean error.

4.5 The effect across 22 properties in two databases

C2DB. Table V and Fig. 6 give the panel. **H1 holds:** all twelve MAE ratios exceed 1.05 (range 1.12–1.92), every interval lies above 1, and every R_{fam} is below 1. For every property the family axis removes more skill than the chemical-system axis. Chemistry is not irrelevant for electronic properties—chemical-

system ratios are 1.03–1.10 for hform, ehull, both gaps, evac, efermi and vbm—but they are small next to 1.4–1.9 for the family axis and are about 1.00 for Y_{2D} , polarisability, plasma frequency, effective mass and v. Stiffness sits mid-panel; evac is the most affected, and the Poisson ratio keeps the least skill (R_{fam} 0.34). Formation energy keeps the most (0.90).

TABLE V — C2DB PANEL (LIGHTGBM, 141 FEATURES, 3 SEEDS, CLUSTER-BOOTSTRAP 95 % CI)

Property	n	Families	MAE ratio, family / random	Skill retained R_{fam}	Ratio, chem. sys.	D_{fam}
hform	16,905	1,198	1.49 [1.43, 1.56]	0.90 [0.89, 0.91]	1.07	0.13
ehull	16,905	1,198	1.48 [1.41, 1.55]	0.63 [0.58, 0.68]	1.03	0.18
gap (PBE)	8,699	753	1.53 [1.41, 1.64]	0.79 [0.75, 0.83]	1.07	0.29
gap (HSE)	3,363	414	1.42 [1.34, 1.51]	0.72 [0.67, 0.78]	1.06	0.21
evac	8,699	753	1.92 [1.77, 2.08]	0.78 [0.73, 0.82]	1.03	0.47
efermi	8,699	753	1.46 [1.39, 1.54]	0.76 [0.73, 0.80]	1.08	0.19
vbm	4,567	566	1.53 [1.42, 1.67]	0.75 [0.69, 0.79]	1.10	0.19
alphax_el	5,352	410	1.12 [1.09, 1.16]	0.83 [0.74, 0.90]	0.99	0.09
plasmafrequency_x	3,931	228	1.55 [1.43, 1.69]	0.61 [0.51, 0.70]	0.99	0.19
emass_cbm	3,532	433	1.23 [1.16, 1.31]	0.68 [0.57, 0.78]	1.00	0.22
Y_{2D}	7,258	676	1.43 [1.29, 1.62]	0.81 [0.71, 0.88]	1.00	0.39
Poisson v	7,229	675	1.28 [1.24, 1.32]	0.34 [0.21, 0.42]	0.99	0.15

Ratio = $MAE_{grouped}/MAE_{random}$. R = skill retained, Eq. (12). D_{fam} = family-mean skill under random splits, Eq. (13).

JARVIS-DFT 2D replication. Table VI lists the ten JARVIS properties. Every interval lies above 1 (ratios 1.03–1.42), so the sign of the effect replicates for all ten. But only **seven of ten exceed 1.05**; the three that do not are the transport properties n-power factor (1.04), n-conductivity (1.03) and n- κ (1.03), which retain 94–96 % of their skill and are poorly predicted under any split. Prediction **P1 therefore failed** (70 % against the required 80 %; all ten intervals above 1 meets its second part). By the pre-

specified decision rule the claim is softened: the penalty is present in every one of the 22 properties tested and above the materiality threshold in 19 of 22, and it is strongest for structure-determined properties. Chemistry matters more in JARVIS than in C2DB: chemical-system ratios reach 1.25 (formation energy), 1.20 (gap) and 1.21 (ϵ_x), although P2 held at its limit (7 of 10 below 1.15). We did not test why; JARVIS is smaller (1,103

versus 16,905 materials) and its family is defined by the space group, not the layer group. Effective mass is the one property classed as chemistry-driven, and its random-split skill is negative (heavy-tailed target, kurtosis 549), so its retention ratio is not

interpretable. P5 held: formation energy shows the penalty in both databases (1.49 and 1.42), as does the band gap (1.53 PBE, 1.35 OptB88vdW).

TABLE VI — JARVIS-DFT 2D REPLICATION (SAME PROTOCOL; FAMILY = ANONYMOUS FORMULA + SPACE GROUP)

Property	n	Families	MAE ratio, family / random	Skill retained R_{fam}	Ratio, chem. sys.	D_{fam}	Dominant axis
formation energy	1,103	302	1.42 [1.30, 1.54]	0.87 [0.83, 0.91]	1.25	0.19	family
gap (OptB88vdW)	1,103	302	1.35 [1.25, 1.45]	0.69 [0.63, 0.75]	1.20	0.15	family
ϵ_x (epsx)	887	261	1.40 [1.23, 1.61]	0.68 [0.58, 0.80]	1.21	0.16	family
avg_elec_mass	678	212	1.23 [1.11, 1.39]	—	1.12	0.23	chemistry
n-Seebeck	806	252	1.22 [1.14, 1.28]	0.31 [0.12, 0.44]	1.09	0.05	family
n-powerfact	802	250	1.04 [1.01, 1.29]	0.94 [0.84, 0.99]	1.02	-0.05	similar
ncond	802	250	1.03 [1.01, 1.39]	0.96 [0.89, 1.00]	1.02	-0.03	similar
nkappa	802	250	1.03 [1.01, 1.50]	0.96 [0.87, 1.02]	1.01	-0.03	similar
exfoliation energy	748	245	1.13 [1.06, 1.21]	0.84 [0.75, 0.92]	1.08	0.18	family
spillage	603	202	1.18 [1.05, 1.39]	—	1.03	0.02	family

Skill retention is shown only where random-split skill exceeds 0.2 (it is negative for effective mass and 0.18 for spillage; spillage R_{fam} is 0.17 [-1.67, 0.80]).

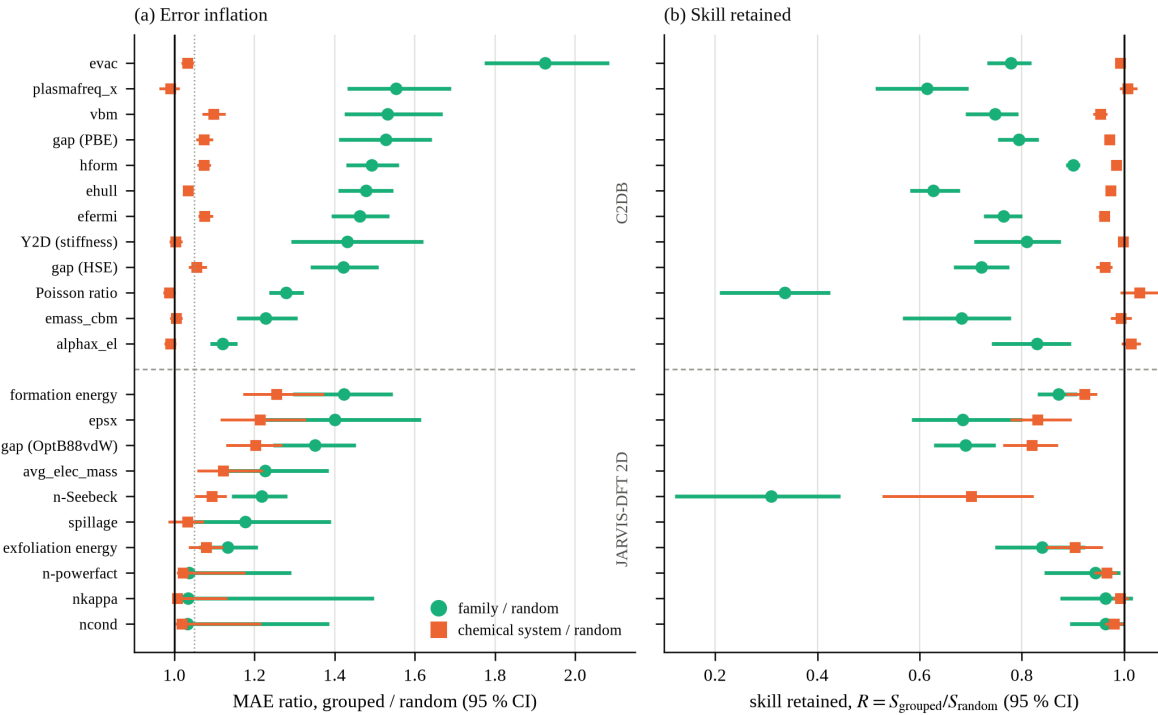


Fig. 6. The family penalty across 22 properties: C2DB (12, lower block) and JARVIS-DFT 2D (10, upper block). (a) Error inflation under family (green) and chemical-system (orange) grouping; dotted line: the 1.05 materiality threshold. (b) Fraction of skill retained (properties with random-split skill below 0.2 omitted). Bars: 95 % cluster-bootstrap intervals. Every green interval in (a) lies above 1; the three JARVIS transport properties sit at the threshold.

4.6 Model class and tuning do not remove the gap

On identical folds the seed-42 CGCNN gives MAE 15.3, 15.9 and 21.1 N/m under random, chemical-system and family splits, against 13.7, 13.9 and 19.8 for LightGBM (Fig. 7). Over three seeds its family/random ratio is 1.38, 1.43 and 1.34 (s.d. 0.03), within the interval of LightGBM’s 1.44 [1.29, 1.62]. The three-seed prediction ensemble reaches MAE 13.70, 14.39 and 19.72 N/m and a ratio of 1.44 [1.33, 1.55]; the difference to LightGBM is -0.003 [-0.10, +0.10] and the paired family-split MAE difference LightGBM - ensemble is +0.09 [-1.60, +1.84] N/m. The seed-42 CGCNN was worse than LightGBM on random and chemical-system splits (paired differences -1.57 and -1.94 N/m, intervals excluding zero) but not on families; the

three-seed ensemble is statistically indistinguishable from LightGBM on random and family splits and does not narrow the gap (A1 and A2 held, A3 did not). Early stopping with grouped validation stopped after 46 epochs on average for family folds versus 122 for random folds. The network is untuned, uses hand-chosen node features and 7,258 training graphs, so a stronger graph model is not excluded.

Nested tuning lowered the Y_{2D} random-split MAE from 13.74 to 12.86 N/m (gain 0.87 [0.55, 1.26]) but left the family-split MAE unchanged, 19.82 to 19.83 (gain -0.01 [-0.28, +0.25]); the ratio rose to 1.54 [1.37, 1.77]. Tuning improved only the optimistic number. For BiDB binding energy (default plus 12 configurations; resampling unit = family, the conservative

choice) tuned MAE is 2.33, 8.83 and 13.75 meV/Å² against 2.43, 9.37 and 13.80 untuned (random, monolayer-grouped, family-grouped); no gain has an interval excluding zero (+0.09 [−0.01,

0.24], +0.53 [−0.02, 1.15], +0.05 [−0.41, 0.52]), and the ratios monolayer/random 3.78 [2.57, 5.66] and family/random 5.89 [3.99, 8.73] are the same as untuned (3.86 and 5.68).

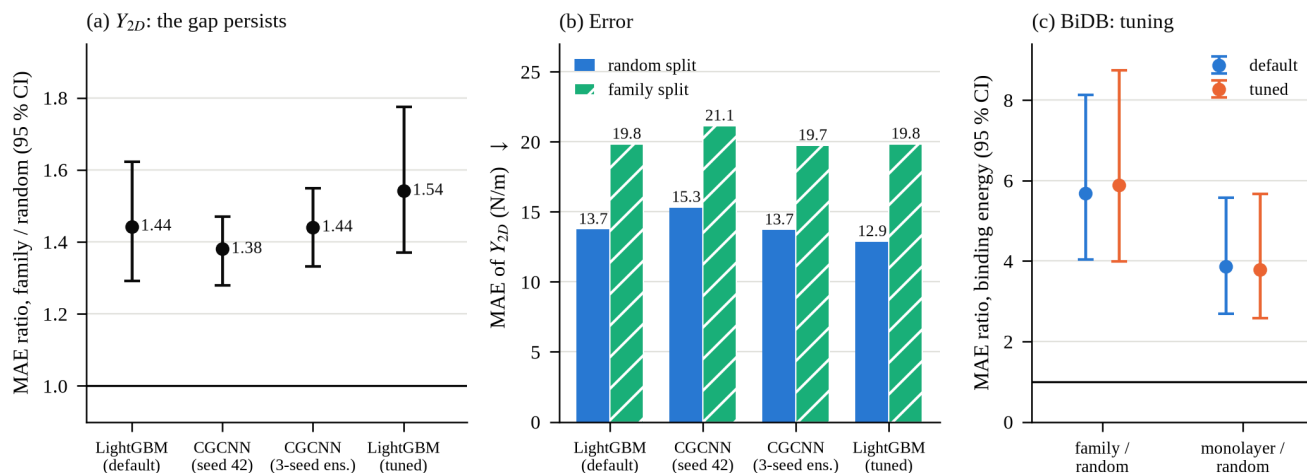


Fig. 7. (a) Y_{2D} family/random MAE ratio with 95 % cluster-bootstrap intervals for default LightGBM, a seed-42 CGCNN, the three-seed CGCNN ensemble and nested-tuned LightGBM. (b) Corresponding MAE under random and family splits. (c) BiDB binding energy: grouped/random ratios with default and nested-tuned LightGBM.

4.7 A label-only forecast of the loss fails

If the loss of skill could be predicted from labels alone, it would be a cheap pre-modelling diagnostic. It cannot (Table VII). In C2DB, H2 failed its own falsification rule ($p = 0.00$, interval -0.60 to 0.81 ; leave-one-property-out error ratio 1.01); the chemical-system analogue ($p = -0.53$ [−0.92, +0.16], error ratio 0.88) points in the hypothesised direction but meets neither criterion; ICC, sample size, family count and kurtosis also do not explain R_{fam} ($|p| \leq 0.44$, all intervals include 0). The prospective replication predicted the same failure and found it: $p = 0.01$ [−0.58, 0.87] in JARVIS (error ratio 1.27) and -0.04 [−0.51, 0.51] pooled over 22 properties (P3 and P4 passed). With ten to twenty-two properties the test has low power, so the safe reading is “no evidence of forecasting”, not “impossible”.

One *exploratory* association was not the pre-specified test: D_{fam} correlates positively with the family/random MAE ratio (pooled $\rho = 0.67$ [0.28, 0.89]; JARVIS 0.75; C2DB 0.37 [−0.36, 0.84]), i.e. properties that a family label already explains well lose more relative accuracy on unseen families. This is partly expected from the definitions (both depend on between-family variance) and is not the forecast of skill retention that was hypothesised.

TABLE VII — PRE-SPECIFIED TESTS AND OUTCOMES

Test	Observed	Needed
C2DB H1: ratios >1.05; all $R < 1$	12/12; yes	descriptive
C2DB H2 $\rho(D_{fam}, R_{fam})$	0.00 [−0.60, 0.81]	≤ -0.6 , CI < 0
C2DB H3 LOPO error ratio	1.01	< 0.8
C2DB H4 chem. system p	−0.53 [−0.92, 0.16]	≤ -0.6
JARVIS P1 ratio >1.05 / CI > 1	7/10 (70 %) / 10/10	$\geq 80\%$ / $\geq 70\%$
JARVIS P2 chem. ratio <1.15	7/10	$\geq 70\%$
JARVIS P3 $\rho(D_{fam}, R_{fam})$	0.01 [−0.58, 0.87]	not ≤ -0.6
Pooled P4 ρ (22 properties)	−0.04 [−0.51, 0.51]	not ≤ -0.6
P5 hform and gap penalty in both DBs	1.49/1.42; 1.53/1.35	both >1.05

P1 failed on its first part; P2–P5 held. For H2–H4 and P3–P4 “held” means the diagnostic did *not* forecast.

4.8 Bilayer binding energy: leakage at the monolayer level

In BiDB each monolayer appears about ten times (one per stacking) with identical monolayer features, and 99.3 % of random-split test bilayers have their monolayer in training. The share of ln-variance between monolayers is 0.955 for binding energy (0.771 for the gap), which caps any monolayer-only model; the random-split $\ln R^2$ (0.940) sits at that ceiling, i.e. the model has learned to recognise monolayers. Fig. 8 and Table VIII give the consequence. LightGBM on *mono_stiffness* has binding-energy MAE 2.43 (random), 9.37 (monolayer-grouped) and 13.80 meV/Å² (family-grouped) at seed 42; the ratios are 3.86 [3.03, 4.80] and 5.68 [4.02, 7.92], and the family-grouped pooled R^2 is 0.15 [−0.09, 0.33] (the mean predictor has MAE 18.5). Across five seeds the grouped MAEs are 9.07 and 13.59, and one family fold (MAE 22.1 against 21.7 for the mean predictor) is worse than the mean. The interlayer gap is only moderately predictable from the monolayer (ratios 1.09 [1.04, 1.15] and 1.29 [1.16, 1.52]).

Two further results qualify what the monolayer information contains. **C2DB monolayer stiffness added no detectable benefit** (binding energy, five seeds, *mono_basic* vs *mono_stiffness*: 2.44 vs 2.42, 9.06 vs 9.07, 13.54 vs 13.59 meV/Å²). **Stacking descriptors** (seed 42 only) helped the gap substantially (LightGBM MAE 0.25 → 0.15 Å random, 0.27 → 0.20 monolayer-grouped, 0.32 → 0.27 family-grouped) but not binding energy for new families (13.85 → 13.92). Extreme z-scan values (up to 7,970 meV/Å²) make the validity filter necessary; the gap results depend on the outlier rule.

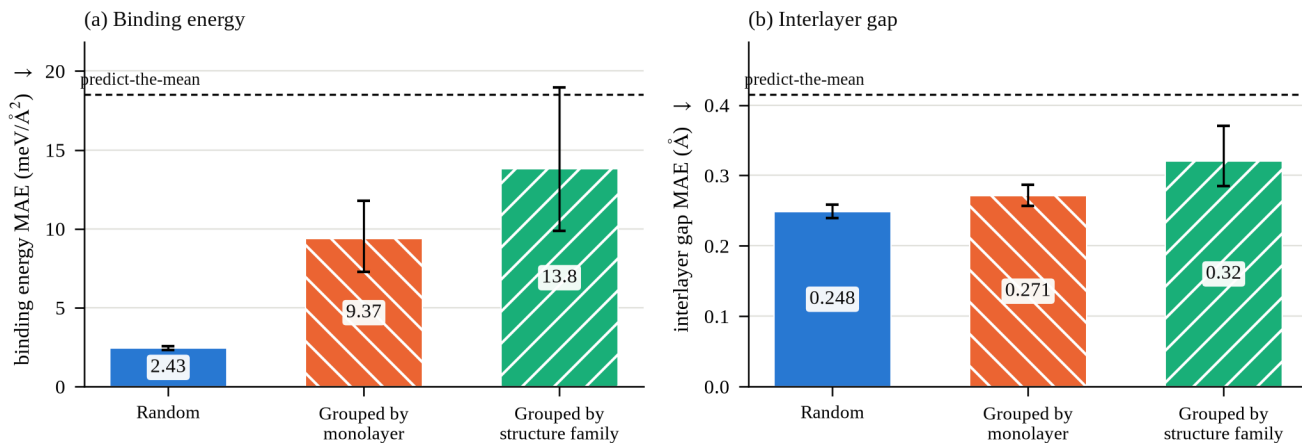


Fig. 8. BiDB bilayer labels predicted from monolayer information (LightGBM, *mono_stiffness*, seed 42; bars: pooled out-of-fold MAE with 95 % cluster-bootstrap intervals; dashed: predict-the-mean). (a) Binding energy. (b) Interlayer gap.

TABLE VIII — BiDB, LIGHTGBM ON *MONO_STIFFNESS*

Label	Scheme	MAE, 5 seeds	MAE, seed 42 [95 % CI]	Ratio to random [95 % CI]	MAE of mean predictor
Binding energy (meV/Å ²)	Random	2.42	2.43 [2.30, 2.58]	1	18.5
	Monolayer	9.07	9.37 [7.27, 11.76]	3.86 [3.03, 4.80]	18.5
	Family	13.59	13.80 [9.86, 18.96]	5.68 [4.02, 7.92]	18.5
Interlayer gap (Å)	Random	0.248	0.248 [0.239, 0.259]	1	0.415
	Monolayer	0.278	0.271 [0.257, 0.287]	1.09 [1.04, 1.15]	0.415
	Family	0.316	0.320 [0.285, 0.370]	1.29 [1.16, 1.52]	0.415

4.9 Cross-database test on JARVIS

Fig. 9 and Table IX report the external test. On all 186 JARVIS rows LightGBM reaches MAE 19.0 [14.4, 24.5] N/m with Spearman 0.82 [0.72, 0.90]. On the 106 matched rows the error is 10.2 [7.9, 12.5] with Spearman 0.93—optimistic, since the material is in training. On the 55 unseen-chemistry rows it is 29.1 [20.5, 39.6] with Spearman 0.73 [0.51, 0.89] and $R^2(\ln)$ 0.45 [-0.14, 0.75]; the lower end shows the sample cannot rule out performance no better than the mean. The “all” row mixes leaky and non-leaky rows and is not a generalisation figure. Composition-only features collapse on unseen chemistry ($R^2(\ln)$

0.06) while structure-only features hold (0.41), but the intervals overlap.

Interpretation. C2DB and JARVIS differ in code, functional and workflow, so even a perfect copy of C2DB would miss JARVIS by about 8.6 N/m. On matched rows the model is close to that floor (10.2); on unseen chemistry its error is about 3.4 times the floor, so most of it is genuine model error, though the floor was measured on matched (possibly easier) materials. One JARVIS reference (HgI_2 , 386 N/m) looks implausible. The JARVIS unit is inferred (median ratio 1.04 to C2DB). The model is useful for *ranking* candidates, not for quoting absolute stiffness on new chemistry.

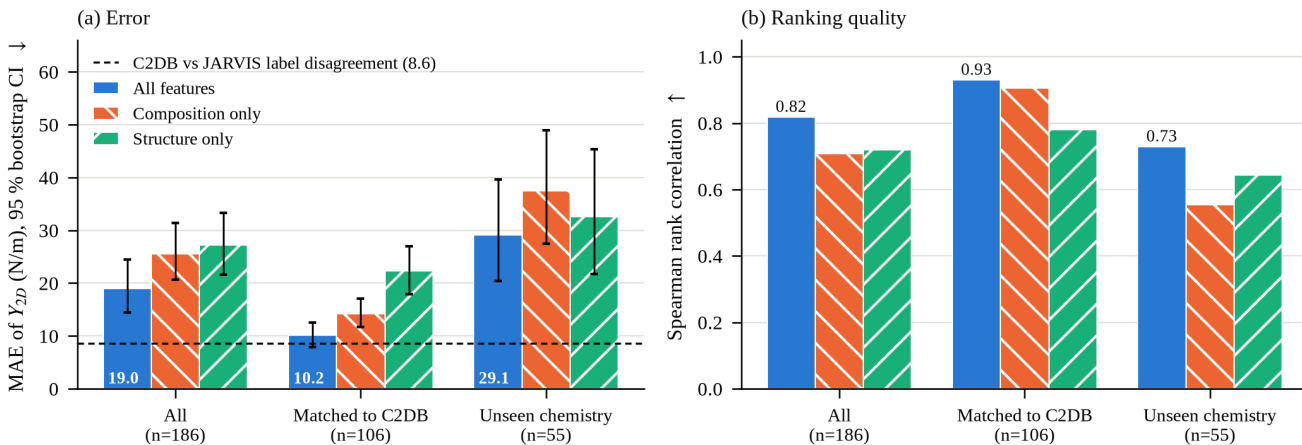


Fig. 9. LightGBM trained on C2DB, tested on JARVIS-DFT. (a) MAE with 95 % bootstrap intervals; dashed: disagreement between the two databases’ own labels on 106 shared materials. (b) Spearman correlation.

TABLE IX — JARVIS EXTERNAL TEST (LIGHTGBM, ALL FEATURES)

Subset	n	MAE	$R^2(\ln)$	Spearman
All	186	19.0	0.55	0.82
Matched to C2DB	106	10.2	0.81	0.93
Unseen chemistry	55	29.1	0.45	0.73
Label agreement	106	8.6	0.75	0.92

4.10 Potentials for monolayer stiffness

Table X and Fig. 10 apply the gate of Eq. (15). MACE-MP-0 and CHGNet fail everywhere and are systematically too soft (pooled slope through the origin 0.62 and 0.48; median predicted/DFT ratio 0.72 and 0.53). MACE-MPA-0 passes only on C2DB (150 monolayers: Spearman 0.94, MAE 9.9 N/m); on JARVIS it has

Spearman 0.81 but MAE 77.5, dominated by a few ML-unstable tensors whose Y_{2D} diverges as $C_{22} \rightarrow 0$ (FeO₂: -10,968 N/m); with those removed (a sensitivity check, not the gate) it gives MAE 9.6/16.2/13.3 (C2DB/JARVIS/combined), narrowly missing both thresholds on the combined set. Part of the JARVIS error is reference noise. The remaining MACE-MPA-0 error is not a geometry effect: on the unchanged C2DB cell the MAE is 9.4 versus 9.9 N/m, and the fractional geometry contribution is 0.05 [-0.09, 0.18], which includes zero. The roughly 11 % softening is unchanged (slope 0.886 vs 0.878); the residual error is material-specific scatter (ratio IQR 0.77–1.10). No potential qualifies as a validated stiffness label generator.

TABLE X — POTENTIAL VS DFT STIFFNESS (GATE: $p \geq 0.90$ AND MAE ≤ 12.93 N/m)

Model	Subset	n	ρ	MAE	Gate
MACE-MP-0	C2DB / JARVIS	150 / 184	0.83 / 0.76	19.3 / 35.4	fail
MACE-MPA-0	C2DB	150	0.94	9.9	pass
	JARVIS / combined	184 / 334	0.81 / 0.87	77.5 / 47.1	fail
CHGNet 0.3.0	C2DB / JARVIS	150 / 184	0.83 / 0.71	23.4 / 42.4	fail

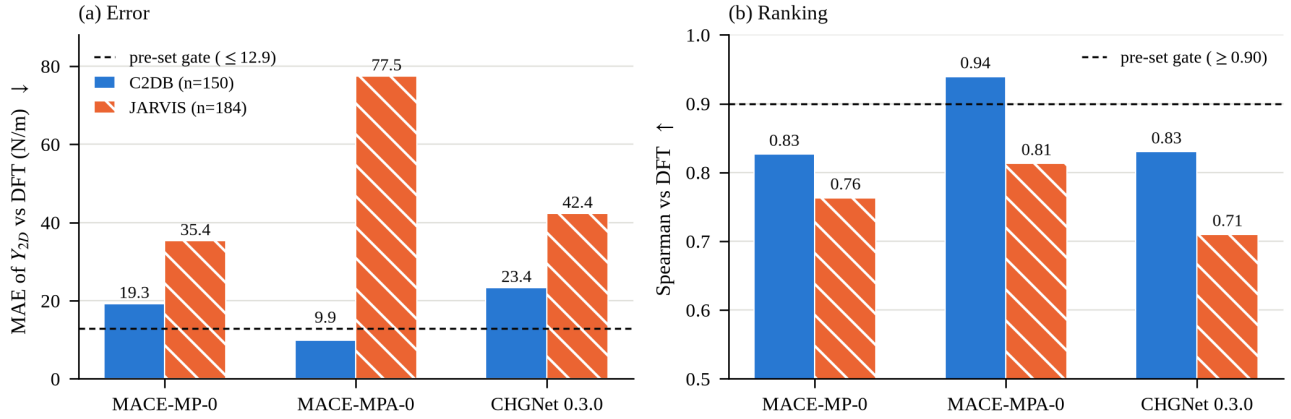


Fig. 10. Potentials against DFT monolayer stiffness. (a) MAE and (b) Spearman by model and subset, with the pre-set gate thresholds (dashed).

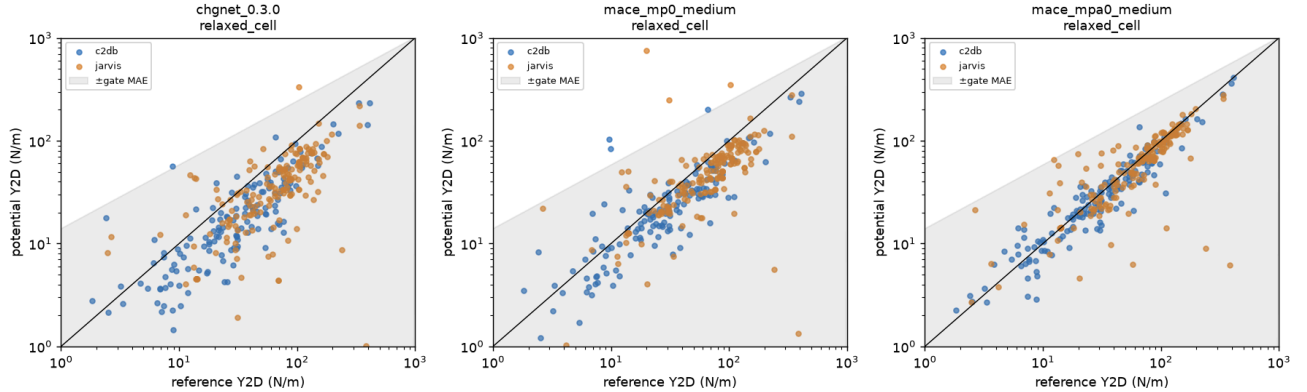


Fig. 11. Parity plots (log-log) of potential-derived against reference Y_{2D} for C2DB (blue) and JARVIS (orange); grey band: gate MAE. Points below the diagonal reveal the softening of MACE-MP-0 and CHGNet.

4.11 Potentials for interlayer binding and bilayer stiffness

Dispersion is essential. Against 40 JARVIS-referenced monolayers (exfoliation energy converted to $\text{meV}/\text{\AA}^2$), MACE-MPA-0 with D3 has a median ratio to the reference of 1.09 [0.88, 1.27] in the van der Waals regime but fails the within-30 % (46 % vs 70 % needed) and failure-rate (16 % vs 10 %) criteria;

without D3 the median ratio is 0.15 [0.09, 0.22], 46 % of structures fail, and D3 supplies a median $19 \text{ meV}/\text{\AA}^2$.

BiDB comparison (307 bilayers, 30 monolayers, D3 on). Binding-energy ranking, magnitude and stacking order pass; the interlayer gap fails (Table XI, Fig. 12). The potential over-binds by about 10 % (bias +1.6, MAE $3.6 \text{ meV}/\text{\AA}^2$); the worst gap

cases are four C_2H_2 bilayers (BiDB gap about 2.0 Å, MACE about 4.1 Å). Pre-specified verdicts “binding energy and distance reproduced” and “stacking dependence reproduced” are therefore **not satisfied**, although S1 passes.

Bilayer stiffness. For 60 homobilayers the ratio of Eq. (17) has a median of 2.008 [1.95, 2.03] (51 strict cases: 2.009), IQR 1.92–2.07, 82 % within [1.8, 2.2] (Fig. 13). In this potential, at small strain, stiffness is thus consistent with *additive layers*. The

variation across stackings of one monolayer is small (median $|R_{\text{best}} - R_{\text{worst}}| = 0.085$, about 4 N/m), and R is only weakly related to binding energy (Spearman 0.14–0.18). Outliers are kept (SnSe 3.3–3.4; PbSe₂ unstable). This is a ratio of two calculations by the same model, **not a DFT validation**, and a pilot at scale was not run because stacking-dependent variation of about 4 N/m could not be separated from model uncertainty without a second potential or DFT.

TABLE XI — MACE-MPA-0 + D3 vs BiDB (PRE-SPECIFIED CRITERIA)

Criterion	Requirement	Result	Verdict
D1 gap	median $ \Delta d \leq 0.10$ Å; $p_{90} \leq 0.30$ Å	0.175 [0.126, 0.275]; 0.479 [0.320, 0.565]	fail
E1 ranking	Spearman ≥ 0.80	0.903 [0.788, 0.950]	pass
E2 magnitude	median ratio 0.8–1.25; ≥ 70 % within ± 30 %; ≥ 90 % within $\times 2$	1.097 [1.037, 1.208]; 72.6 %; 98.4 %	pass
E3 failures	≤ 10 %	2 of 307 (0.7 %)	pass
S1 stacking	within-monolayer $\rho \geq 0.60$; best in top 2 for ≥ 60 %	0.749 [0.665, 0.847]; 70 %	pass

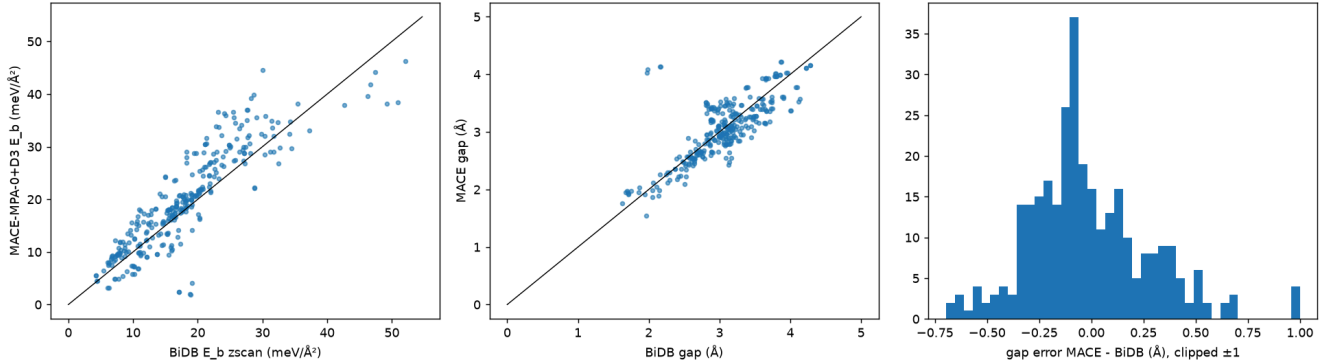


Fig. 12. MACE-MPA-0 + D3 against BiDB for 307 bilayers: binding energy (left), interlayer gap (middle), gap error (right).

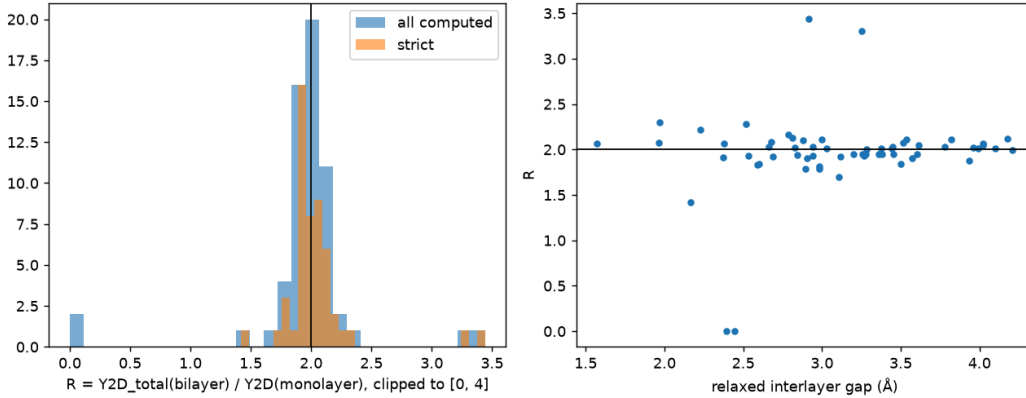


Fig. 13. Ratio of bilayer to monolayer in-plane stiffness for 60 homobilayers (MACE-MPA-0 + D3, small strain). Left: distribution (all computed and strict subsets). Right: ratio against relaxed interlayer gap. A ratio of 2 (black line) corresponds to additive layers. Not DFT-validated.

5 DISCUSSION

Our central finding is methodological: in these data, accuracy depends far more on *what the test set shares with the training set* than on the regressor. The two tree ensembles differ by about 1 N/m, a three-seed graph-network ensemble matches gradient boosting and nested tuning does not remove the family penalty, whereas changing the split changes the MAE of the same model by about 6 N/m for stiffness and by a factor 3.9–5.7 for bilayer binding energy. Reporting a random-split number alone would have hidden this. The model nevertheless retains real skill on unfamiliar families (MAE 19.9 vs 42.4 for the mean predictor)

and ranks JARVIS materials reasonably, which is what a screening tool needs.

Three results deserve emphasis. First, chemical-system grouping alone would have missed the effect for stiffness (ratio 1.00), although it costs a small amount for several electronic properties. Second, the family penalty is present for every one of the 22 property–database pairs tested and above the 1.05 threshold for 19, but how large it is cannot be forecast from labels, and the failed P1 shows it is not uniformly large: the three JARVIS transport properties, which are poorly predicted under any split, barely lose accuracy, and chemistry matters more in JARVIS than in C2DB. Third, universal potentials are only partially

trustworthy: the best monolayer result (MACE-MPA-0 on C2DB) does not carry over to JARVIS, dispersion is essential for interlayer binding, and the interlayer gap fails the criterion. Because the potentials were pretrained on large DFT collections of bulk crystals and some C2DB monolayers derive from layered bulk compounds, overlap between their training data and our benchmarks has not been assessed.

5.1 Limitations

- One training database and DFT workflow (C2DB, GPAW/PBE); a small external set (186 rows, 55 unseen-chemistry) with wide intervals; JARVIS units inferred and C2DB-JARVIS matching heuristic.
- “Family” is our own definition; other definitions could change the gap. Family sizes are uneven (up to 432 rows; 1,148 bilayers in the largest BiDB family), so folds are lumpy. The mechanism (symmetry-determined structure-property maps versus how databases are built by element substitution) is not established.
- Two databases, but the JARVIS replication uses a coarser family (anonymous formula plus space group) and about 1,100 materials; the graph network is untuned with hand-chosen node features (three seeds); tuning searched 16 (Y_{2D}) and 12 (binding energy) random configurations; the panels use default LightGBM; ten to 22 properties give limited power for the label-only test.
- Polymorphs are retained, only stable materials are modelled (14 % of tensor-bearing entries removed), and ablations are group-level with a single model.
- BiDB labels come from one workflow (PBE-D3, rigid layers, homobilayers); the uid map is verified at formula level only; the g_s binding-energy column has no available definition and is not used; tuning was run for the binding energy only and no graph network was run for Task B.
- Potential checks use small samples (150 + 184 monolayers; 30 for BiDB), one passing model, exclude magnetic and

strongly bonded layers, and report no DFT validation of any bilayer stiffness.

5.2 Future work

Not yet done: a mechanistic analysis of why prototype overlap, rather than chemistry, carries the leakage; datasets beyond 2D materials; a graph network for the BiDB targets; and, before any claim on bilayer stiffness, a DFT reference on a few bilayers (including SnSe and BaBr₂), an independent second potential to test cancellation in the ratio of Eq. (17), an explanation of the gap error, and treatment of magnetic and metallic layers.

6 CONCLUSION

With 141 DFT-free descriptors, LightGBM predicts the 2D Young’s modulus of C2DB monolayers with an MAE of 13.8 N/m under random splitting but 19.9 N/m (1.44×) when structure families are held out; holding out chemical systems costs nothing. The family penalty is present for all 22 properties tested in two databases (above the materiality threshold for 19), survives a graph-network ensemble and tuning, and cannot be forecast from labels; for bilayer binding energies leakage at the monolayer level inflates random-split accuracy almost four-fold. On an independent database the model ranks materials well but is only roughly quantitative on new chemistry, and no universal potential yet qualifies as a stiffness-label generator. Family-grouped evaluation should therefore be the default for 2D-materials benchmarks, and bilayer stiffness remains an open, unvalidated question.

7 DATASETS, LICENCES AND CREDITS

Raw databases are not redistributed; only derived statistics, figures and code are. Table XII lists each resource, its role and how it must be credited. *Licence terms below were taken from the providers’ published statements as recalled at drafting time; they could not be re-checked from the drafting environment, so please confirm each entry on the provider’s site before submission.*

TABLE XII — DATA RESOURCES, LICENCES AND REQUIRED CREDIT

Resource	Role in this work	Provider / how to credit	Licence / terms (to be confirmed)
C2DB Computational 2D Materials Database	Training and cross-validation data; stiffness tensors (GPAW/PBE), structures, layer groups, panel properties. 8,462 stiffness records → 7,258 used; 16,905 entries for the panel.	CAMD, Technical University of Denmark. Cite Hastrup <i>et al.</i> , 2D Mater. 5, 042002 (2018) [15] and Gjerding <i>et al.</i> , 2D Mater. 8, 044002 (2021) [16]. https://c2db.fysik.dtu.dk/	Open data; to our recollection released under Creative Commons Attribution 4.0 (CC BY 4.0), i.e. reuse with attribution. Confirm on the C2DB site.
JARVIS-DFT (2D subset)	External test, 10-property panel replication (1,103 monolayers) and potential references (never used to train the C2DB models); stiffness tensors, exfoliation energies and other properties (VASP/OptB88vdW). 229 tensors → 186 used.	NIST (Choudhary, Tavazza and co-workers). Cite Choudhary <i>et al.</i> , npj Comput. Mater. 6, 173 (2020) [19] and Phys. Rev. B 98, 014107 (2018) [20]. https://jarvis-tools.readthedocs.io/	Openly distributed by NIST (data via the JARVIS website / figshare; code under the NIST open-source licence). Attribution required; confirm the exact data licence on the JARVIS / figshare record.
BiDB bilayer database	Task B labels (binding energy, interlayer gap; 10,192 homobilayers of 992 monolayers) and interlayer validation of the potentials.	Cite Pakdel <i>et al.</i> , Nat. Commun. 15, 932 (2024) [23]. https://2dhub.org/bidb/bidb.html	Licence not recorded in the project; obtain and state the terms from the BiDB site before submission.
Other links recorded	HetDB (2dhub.org) and arXiv:2004.03025 appear in the project’s link list. Not used in any reported experiment.	—	Not applicable (not used).

Code and analysis files are released by the authors under the MIT licence as stated in the project README. The licence file referenced there was not present in the repository checkout used for drafting and should be restored before release. Software: Python 3.11, scikit-learn 1.9.1 [27], LightGBM 4.7.0 [28], pymatgen [26], matminer 0.10.1 [25], PyTorch for the CGCNN, ASE 3.29.0 [36], mace-torch 0.3.16 [34], CHGNet 0.4.2 [35], torch-dftd for D3 [37]. Fixed seed 42 (stiffness seeds 42–46; panel 42–44). All numbers and figures are produced from the project result files (mat-ml/results/) by paper/build/make_figures.py and paper/build/build.py.

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REFERENCES

- [1] V. Venturi, H. Parks, Z. Ahmad, and V. Viswanathan, “Machine learning enabled discovery of application dependent design principles for two-dimensional materials,” arXiv:2003.13418, 2020.
- [2] M. Fronzi, M. Abu Ghazaleh, O. Isayev, D. A. Winkler, J. Shapter, and M. J. Ford, “Impressive computational acceleration by using machine learning for 2-dimensional super-lubricant materials discovery,” arXiv:1911.11559, 2020.
- [3] B. Meredig *et al.*, “Can machine learning identify the next high-temperature superconductor? Examining extrapolation performance for materials discovery,” *Mol. Syst. Des. Eng.*, vol. 3, pp. 819–825, 2018.
- [4] D. R. Roberts *et al.*, “Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure,” *Ecography*, vol. 40, pp. 913–929, 2017.
- [5] K. Li, D. Persaud, K. Choudhary, B. DeCost, M. Greenwood, and J. Hattrick-Simpers, “Exploiting redundancy in large materials datasets for efficient machine learning with less data,” *Nat. Commun.*, vol. 14, p. 7283, 2023.
- [6] Q. Li, N. Fu, S. S. Ome, and J. Hu, “MD-HIT: Machine learning for material property prediction with dataset redundancy control,” *npj Comput. Mater.*, 2024.
- [7] “MatFold: systematic insights into materials discovery models’ performance through standardized cross-validation protocols,” software and preprint, <https://github.com/d2r2group/MatFold> (authors to be added).
- [8] D. Davidovikj *et al.*, “Young’s modulus of 2D materials extracted from their nonlinear dynamic response,” arXiv:1704.05433, 2017.
- [9] R. Zhang and R. Cheung, “Mechanical properties and applications of two-dimensional materials,” book chapter, InTech, 2016.
- [10] R. M. Elder, M. R. Neupane, and T. L. Chantawansri, “Mechanical properties of homogeneous and heterogeneous layered 2D materials,” in *Proc. SISPAD* (IEEE).
- [11] P. Malakar, M. S. H. Thakur, S. M. Nahid, and M. M. Islam, “Data-driven machine learning to predict mechanical properties of monolayer TMDs,” manuscript (venue not stated in the copy reviewed).
- [12] M. Maurizi, C. Gao, and F. Berto, “Predicting stress, strain and deformation fields in materials and structures with graph neural networks,” *Sci. Rep.*, vol. 12, p. 21834, 2022.
- [13] A. Dunn, Q. Wang, A. Ganose, D. Dopp, and A. Jain, “Benchmarking materials property prediction methods: the Matbench test set and Automatminer reference algorithm,” *npj Comput. Mater.*, vol. 6, p. 138, 2020.
- [14] BiMat-ML (band-gap prediction on BiDB), arXiv:2606.01012 (as recorded in the project notes; authors and title to be completed).
- [15] S. Hastrup *et al.*, “The Computational 2D Materials Database: high-throughput modeling and discovery of atomically thin crystals,” *2D Mater.*, vol. 5, no. 4, p. 042002, 2018.
- [16] M. N. Gjerding *et al.*, “Recent progress of the Computational 2D Materials Database (C2DB),” *2D Mater.*, vol. 8, no. 4, p. 044002, 2021.
- [17] J. J. Mortensen, L. B. Hansen, and K. W. Jacobsen, “Real-space grid implementation of the projector augmented wave method,” *Phys. Rev. B*, vol. 71, p. 035109, 2005.
- [18] J. P. Perdew, K. Burke, and M. Ernzerhof, “Generalized gradient approximation made simple,” *Phys. Rev. Lett.*, vol. 77, pp. 3865–3868, 1996.
- [19] K. Choudhary *et al.*, “The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design,” *npj Comput. Mater.*, vol. 6, p. 173, 2020.
- [20] K. Choudhary, G. Cheon, E. Reid, and F. Tavazza, “Elastic properties of bulk and low-dimensional materials using van der Waals density functional,” *Phys. Rev. B*, vol. 98, p. 014107, 2018.
- [21] G. Kresse and J. Furthmüller, “Efficient iterative schemes for *ab initio* total-energy calculations using a plane-wave basis set,” *Phys. Rev. B*, vol. 54, pp. 11169–11186, 1996.
- [22] J. Klimeš, D. R. Bowler, and A. Michaelides, “Van der Waals density functionals applied to solids,” *Phys. Rev. B*, vol. 83, p. 195131, 2011.
- [23] S. Pakdel *et al.*, “High-throughput computational stacking reveals emergent properties in natural van der Waals bilayers,” *Nat. Commun.*, vol. 15, p. 932, 2024.
- [24] L. Ward, A. Agrawal, A. Choudhary, and C. Wolverton, “A general-purpose machine learning framework for predicting properties of inorganic materials,” *npj Comput. Mater.*, vol. 2, p. 16028, 2016.
- [25] L. Ward *et al.*, “Matminer: An open source toolkit for materials data mining,” *Comput. Mater. Sci.*, vol. 152, pp. 60–69, 2018.
- [26] S. P. Ong *et al.*, “Python Materials Genomics (pymatgen): A robust, open-source Python library for materials analysis,” *Comput. Mater. Sci.*, vol. 68, pp. 314–319, 2013.
- [27] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in Python,” *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [28] G. Ke *et al.*, “LightGBM: A highly efficient gradient boosting decision tree,” in *Proc. NeurIPS*, vol. 30, 2017.
- [29] A. E. Hoerl and R. W. Kennard, “Ridge regression: Biased estimation for nonorthogonal problems,” *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.
- [30] L. Breiman, “Random forests,” *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [31] T. Xie and J. C. Grossman, “Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties,” *Phys. Rev. Lett.*, vol. 120, p. 145301, 2018.
- [32] B. Efron, “Bootstrap methods: Another look at the jackknife,” *Ann. Statist.*, vol. 7, no. 1, pp. 1–26, 1979.
- [33] C. Spearman, “The proof and measurement of association between two things,” *Am. J. Psychol.*, vol. 15, no. 1, pp. 72–101, 1904.
- [34] I. Batatia *et al.*, “A foundation model for atomistic materials chemistry,” arXiv:2401.00096, 2023.
- [35] B. Deng *et al.*, “CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling,” *Nat. Mach. Intell.*, vol. 5, pp. 1031–1041, 2023.
- [36] A. H. Larsen *et al.*, “The atomic simulation environment—a Python library for working with atoms,” *J. Phys.: Condens. Matter*, vol. 29, p. 273002, 2017.
- [37] S. Grimme, J. Antony, S. Ehrlich, and H. Krieg, “A consistent and accurate *ab initio* parametrization of density functional dispersion correction (DFT-D) for the 94 elements H–Pu,” *J. Chem. Phys.*, vol. 132, p. 154104, 2010.